PROJECT THREE MILESTONE TWO: DESIGN EXPLORATION AND DESIGN REVIEW #1

MILESTONE 2.1 – RESEARCH ASSIGNMENT

Team ID: Mon-23

Complete this worksheet before Design Studio for Week 7.

- State the question you plan to answer through your research
- Summarize your research findings (answer). Your answer should be a coherent, well-written summary of your research, not a "brain dump".
- You may include images, but don't forget to cite them properly.
- Aim for a length of about 500 words
- Properly cite your sources using IEEE formatted references and in-text citations. For information on referencing formats and choosing sources, see Design and Communication Workshop 1.

Name: Prabhjot Bhandal	MacID: bhandp9

What is your question?

Do previously able-bodied individuals feel less connected to the world after acquiring a disability? What strategies help a previously able-bodied individual adapt to reduced spatial awareness caused by a disability?

What is your answer?

When someone experiences reduced spatial awareness due to a recently acquired disability, it can lead to a feeling of disconnection from the world. This is very true for individuals with disabilities as they are less likely to have close friends, relatives and local contacts [1,2]. Moreover, the loss of spatial awareness makes it more difficult for them to experience positive social interactions, which worsens their mental health [1,2]. To help individuals adapt and navigate their reduced spatial awareness, several strategies are used, including environment modification, sensory integration and mental representation exercises [3,4].

One strategy that is used is to modify the individual's everyday environment. By rearranging furniture or creating wider pathways, individuals can make their homes easier and safer to navigate [3]. Visual cues, like bright colours and distinct lighting, can be used to identify separate locations within a space [3]. Bright lighting can also aid in this by increasing contrast but can also reduce potential hazards and make different objects easier to see [3].

Another strategy that is used involves utilizing all sensory information to compensate for spatial limitations. People with limited spatial awareness can develop techniques to use all their available

senses to understand their surroundings [4]. For example, mobility aids like white canes allow for visually impaired individuals to interpret their environment through tactile feedback and sound, which helps them regain some of the spatial awareness they have lost [5]. This recontextualization of sensory input enables them to navigate effectively without having to only rely on sight [5].

Alternatively, individuals can engage in mental representation exercises, which help improve spatial awareness and fine motor skills [4,6]. Through motor imagery training, individuals mentally simulate physical movements, improving motor planning, coordination, and precision, which are all crucial for those with limited spatial awareness [7]. Basic Action Concepts (BACs) are an extension of motor imagery as they help improve cognitive abilities related to performing complex actions with precision [6,7]. These exercises enable individuals to enhance their spatial awareness and fine-tune their movements without relying on external aids [4,6].

In conclusion, there are various ways to improve spatial awareness and reconnect individuals with their environment. Environmental modifications help reduce hazards and make navigation easier. By utilizing other senses in intuitive ways, with or without aids, individuals can better understand their surroundings. Alternatively, mental exercises can actively improve spatial awareness, enhancing both motor skills and precision. Together, these strategies can help individuals regain a strong sense of connection to their environment.

List of sources:

[1] Y. Strautins and C. McDiarmid, "Social connections among persons with and without mental health-related disabilities, 2020," *www150.statcan.gc.ca*, May 31, 2023. https://www150.statcan.gc.ca/n1/pub/89-654-x/89-654-x2023001-eng.htm

[2] Statistics Canada, "The Daily — Exploring the social interactions of those with and without mental health-related disabilities," *Statcan.gc.ca*, 2023. https://www150.statcan.gc.ca/n1/daily-quotidien/230531/dq230531f-eng.htm

[3] The OT Centre UK, "Private Occupational Therapy and Sensory Integration | The OT Centre - Spatial Awareness Deficits," *Theotcentre.co.uk*, 2024. https://www.theotcentre.co.uk/difficulties-treated-with-occupational-therapy-and-si-therapy/spatial-awareness-deficits

[4] J. Duquette and F. Baril, "Spatial orientation in adolescents with visual impairment: related factors and avenues for assessment," Mar. 2012. Available: https://extranet.inlb.qc.ca/wp-content/uploads/2015/01/Spatial-orientation-in-adolescents-with-VI_factors_assessment.pdf

[5] CNIB Foundation, "About the White Cane," *CNIB*. https://www.cnib.ca/en/about-white-cane?region=on

[6] T. Schack, K. Essig, C. Frank, and D. Koester, "Mental representation and motor imagery training," *Frontiers in Human Neuroscience*, vol. 8, May 2014, doi: https://doi.org/10.3389/fnhum.2014.00328.

[7] M. F. Almufareh, S. Kausar, M. Humayun, and S. Tehsin, "Leveraging Motor Imagery Rehabilitation for Individuals with Disabilities: A Review," *Healthcare*, vol. 11, no. 19, p. 2653, Jan. 2023, doi: https://doi.org/10.3390/healthcare11192653.

MILESTONE 2.1 – INITIAL CONCEPT EXPLORATION

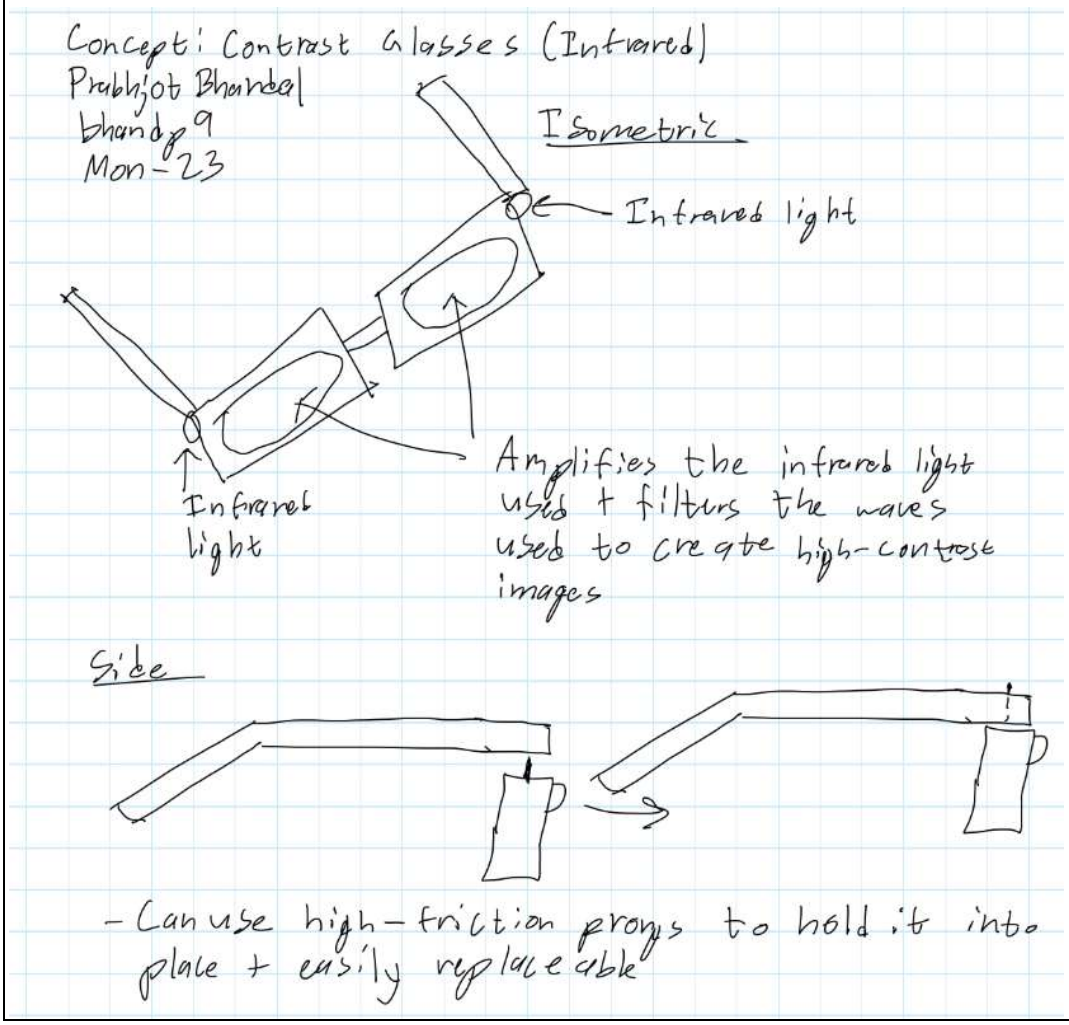
Team ID: Mon-23

Complete this worksheet before Design Studio for Week 7.

1. Include multiple images of your **initial** concept exploration, if needed
   → Include necessary annotations to help in the communication of your ideas
   → These can be photos of hand sketches, photos of initial prototypes, screen grabs of basic CAD models
   → Include your Team Number, Name and MacID on *each* concept image
2. Insert your photo(s) as a Picture (Insert > Picture > This Device)
3. **Do not include more than *two* concept images per page**

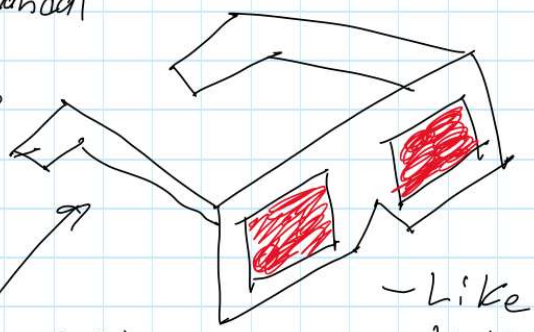
Name: Prabhjot Bhandal	MacID: bhandp9

Insert screenshot(s) of your concept below.



Alternative: Contrast Glasses that work with opposite colours.

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Mon-23





- would be cost-effective; 3D glasses at movie theaters are cheap & easy to make

- Like 3D glasses, can distinctly see objs of opposite colour
  - ↳ issue: would need to get lots of Live objects

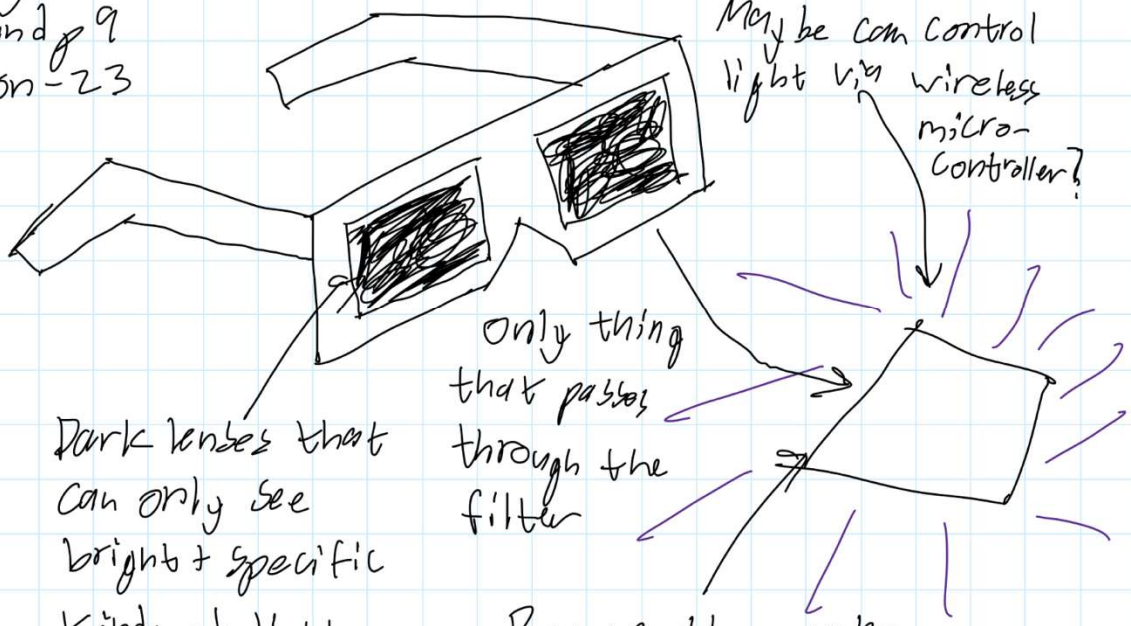
Alternative: Contrast glasses w/ UV light and reuseable markers.

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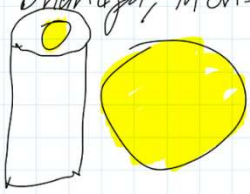
Alternative! Contrast Glasses w/ lenses that point at ~~glowing~~ markers

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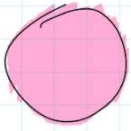


Concept: Audio Sticky Notes

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- Sticks onto anything
- Has a programmable NFC tag
  - ↳ Can be programmed w/ a phone
  - ↳ Text-to-speech audio
- Heat resistant
- High-contrast, neon colour
- Reusable



← can come in diff. colours

Issues:
- Finite
- would need to know what something is before labelling
- Isn't obvious how to use it immediately

Fixes?!
- Make a QR code generator, infinite QR codes
- Give her prompts of how to use it
  - ↳ would be in audio

MILESTONE 2.3 – REFINED CONCEPT EXPLORATION

Team ID: Mon-23

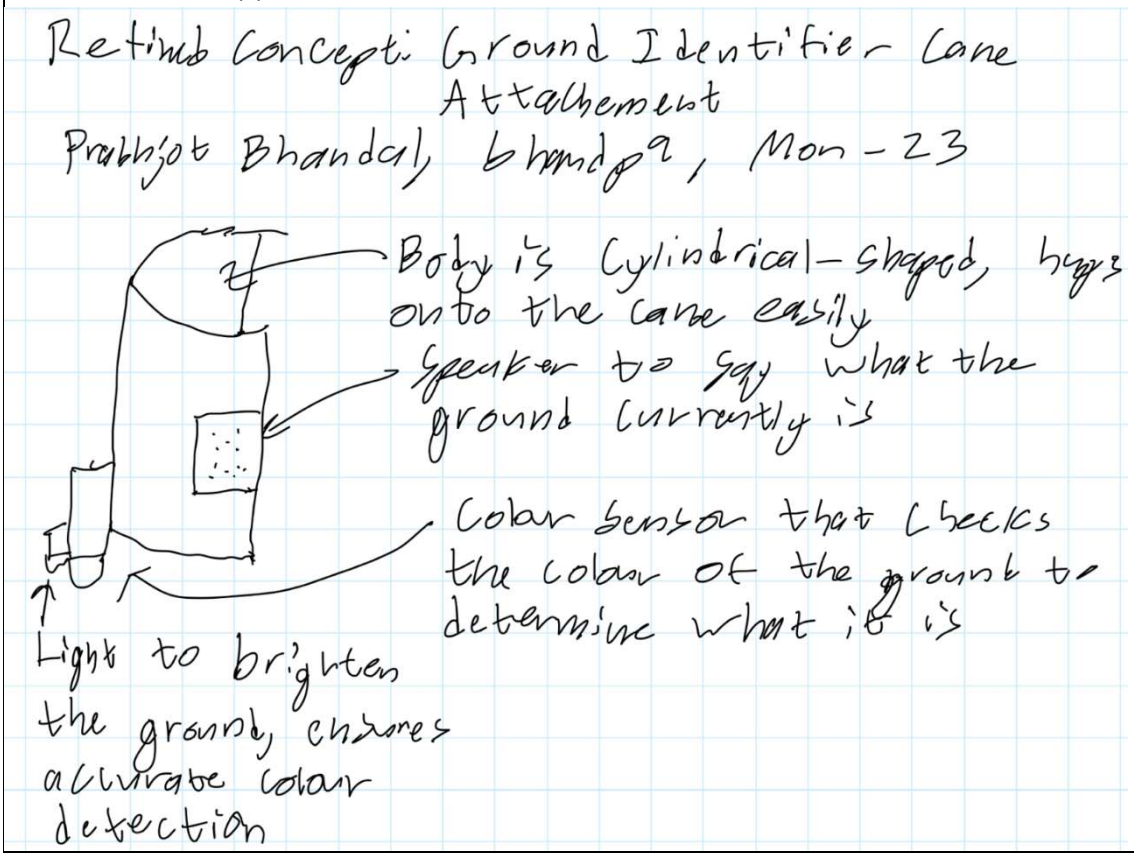
Complete this worksheet during Design Studio for Week 7.

4. Include multiple images of your **refined** concept exploration, if needed
   - → Include 2 distinct concepts based on the functional analysis
   - → Include necessary annotations to help in the communication of your ideas
   - → These can be photos of hand sketches, photos of initial prototypes, screen grabs of basic CAD models
   - → Include your Team Number, Name and MacID on *each* concept image
5. Insert your photo(s) as a Picture (Insert > Picture > This Device)
6. **Do not include more than *two* concept images per page**

Concept 1: Ground Identifier Cane Attachment

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Insert screenshot(s) of your concept below.



Team ID: Mon-23

Concept 2: Neon Audio Labels in Fun Shapes

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Insert screenshot(s) of your concept below.

